

Applied Optimization

Solution Exercise 5 - Line Search Methods

Exact line search for the convex quadratic function (0.5 pt)

Consider the convex quadratic function

$$f(x) = \frac{1}{2}x^T Qx + q^T x + c,$$

where $x \in \mathbb{R}^n$, and constant parameters Q is a symmetric positive definite matrix $\in \mathbb{R}^{n \times n}$, q is a vector in $\mathbb{R}^n$, and $c \in \mathbb{R}$.

Given a search direction Δx , compute the exact line search parameter t for an arbitrary point x in the domain of f .

$$t := \arg \min_{s \geq 0} f(x + s\Delta x)$$

Solution:

let $l(s)$ be our line function

$$l(s) := f(x + s\Delta x),$$

to solve it we compute its critical points, set to zero and solve for s .

$$l'(s) = \nabla f(x + s\Delta x)^T \Delta x = 0,$$

with

$$\nabla f(x) = Qx + q$$

Evaluate at $x + s\Delta x$ and plugged in the equation above:

$$\begin{aligned} (Q(x + s\Delta x) + q)^T \Delta x &= 0 \\ \Delta x^T (Qx + sQ\Delta x + q) &= 0 \\ \Delta x^T Qx + s\Delta x^T Q\Delta x + \Delta x^T q &= 0 \\ s &= -\frac{\Delta x^T Qx + \Delta x^T q}{\Delta x^T Q\Delta x} \end{aligned}$$

Since $s \geq 0$, $t = \max(s, 0)$

Gradient descent with exact line search (0.5 pt)

Consider the unconstrained minimization problem:

$$\text{minimize } \frac{1}{4}x_1^2 + x_2^2$$

starting from point $x^{(0)} = (2, 1)$, perform one iteration of the gradient descent algorithm with exact line search. Sketch the function, the line and the update. ¹

What is the value of $\|\nabla f(x^{(1)})\|^2$ after the update $x^{(1)} := x^{(0)} + t\Delta x$?

Solution:

We use the formulas above. To do this we identify, Q , q and c , using with $x = (x_1, x_2)^T$ by writing the function in matrix form:

$$\begin{aligned}\frac{1}{4}x_1^2 + x_2^2 &= \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + 0 \\ &= \frac{1}{2}x^T Qx + q^T x + c\end{aligned}$$

Thus $c = 0$, $q = (0, 0)^T$ and

$$\frac{1}{2}Q = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix}$$

For the computations we need the following:

Direction: $\Delta x = -\nabla f = -Qx - q$

Step: $t = -\frac{\Delta x^T Qx + \Delta x^T q}{\Delta x^T Q \Delta x} = -\frac{\Delta x^T Qx}{\Delta x^T Q \Delta x}$

Update: $x = x + t\Delta x$

$$\begin{aligned}\Delta x^{(0)} &= -\nabla f(2, 1) = -\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \end{pmatrix} \\ t^{(0)} &= -\frac{\begin{pmatrix} -1 & -2 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix}}{\begin{pmatrix} -1 & -2 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ -2 \end{pmatrix}} = -\frac{\begin{pmatrix} -1 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}}{\begin{pmatrix} -1 & -2 \end{pmatrix} \begin{pmatrix} -1/2 \\ -4 \end{pmatrix}} = -\frac{-5}{\frac{1}{2} + 8} = 10/17 \approx 0.5882 \\ x^{(1)} &= \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \frac{10}{17} \begin{pmatrix} -1 \\ -2 \end{pmatrix} = \frac{1}{17} \begin{pmatrix} 24 \\ -3 \end{pmatrix} \approx \begin{pmatrix} 1.4118 \\ -0.1765 \end{pmatrix} \\ \|\nabla f(x^{(0)})\|^2 &= \left\| \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\|^2 = (1^2 + 2^2) = 5\end{aligned}$$

¹here the link to our interactive plotting system

https://slides.cg.g.unibe.ch/aopt20/plots/plot2d_create.html

$$\|\nabla f(x^{(1)})\|^2 = \left\| \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix} \frac{1}{17} \begin{pmatrix} 24 \\ -3 \end{pmatrix} \right\|^2 = \frac{1}{17^2} (12^2 + 6^2) = \frac{1}{289} (144 + 36) = \frac{180}{289} \approx 0.6228$$

Figure 1 that you could reproduce on our interactive visualization system:

<https://slides.cgg.unibe.ch/aopt20/plots/descent2d.html>

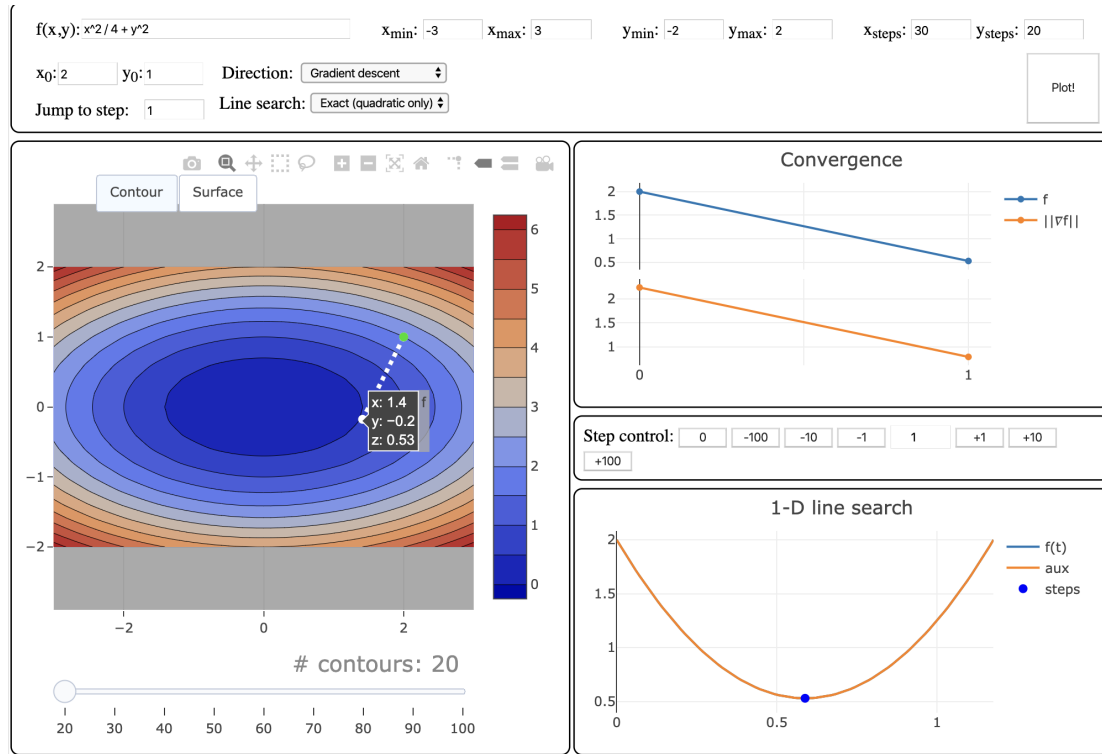


Figure 1: Sketch of the above computations generated from our interactive visualization system